

8×8 PROGRAMMABLE MATRIX DISPLAY

Teaching Aid For Primary Schools

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The custom-built matrix display for primary school children's character visualisation is an accessible innovation that strengthens primary education. Its 8 × 8 programmable format offers a seamless and dynamic platform for young learners to visually recognise and understand alphabets and numbers, supported by user-friendly programmability.

By integrating modern technology into the classroom, the system delivers an economical learning solution without straining resources, owing to its low-cost design. This approach deepens understanding of language and characters while sparking curiosity through an effective blend of technology and education.

The matrix display also paves the way for a more engaging and effective learning experience, helping primary school students progress confidently in character recognition and literacy development.

Transforming the approach to elementary education, the 8 × 8 programmable matrix display serves as an innovative and engaging teaching aid for character visualisation. The system creates a dynamic and visually appealing platform through which children can explore and understand characters more effectively. By seamlessly combining technology with learning, it stimulates curiosity while fostering a deeper understanding of language and letter formation.

As modern educational technologies continue to evolve, this teaching aid paves the way for a more interactive and effective learning experience, supporting primary students in their journey of

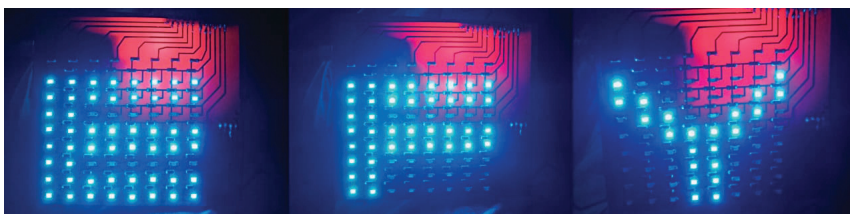


Fig. 1: Authors' prototype along with a sample output displayed as an EFY light illusion on the 8×8 LED matrix

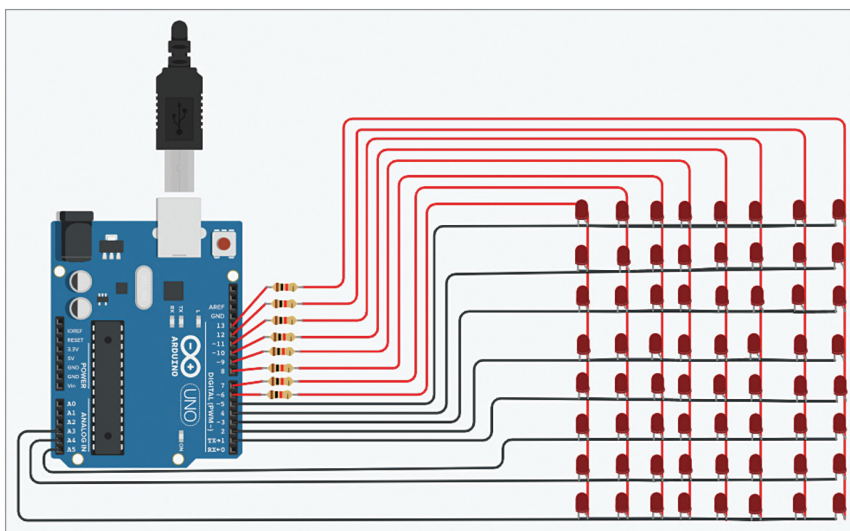


Fig. 2: Circuit connection of 8×8 SMD matrix display interfaced with Arduino Uno

literacy development and character recognition.

This LED matrix can generate multiple patterns using an Arduino program based on high and low logic levels. Fig. 1 shows the finished system along with a sample output displayed as an EFY light illusion on the 8 × 8 LED matrix. The circuit can be powered using a 5V DC supply. The components required for this system are listed in the Bill of Materials table.

Circuit and working

The matrix display is arranged in rows and columns and interfaced with the Arduino Uno. The columns

BILL OF MATERIALS

Name	Specification	Quantity
Arduino Uno	Board	1
SMD LEDs	1206 package	64
SMD resistors	1206, 1kΩ	8
SMD resistors	1206, 0Ω	56
FR4 PCB board	10cm×10cm	1
Male-to-male jumper wires	—	As required
Single-pin connectors	—	As required
Wires, solder wires	Soldering iron	As required

are connected to the cathodes of the LEDs, so a column must be set low to enable an LED, while the rows are connected to the anodes and must be set high to turn on a specific LED.